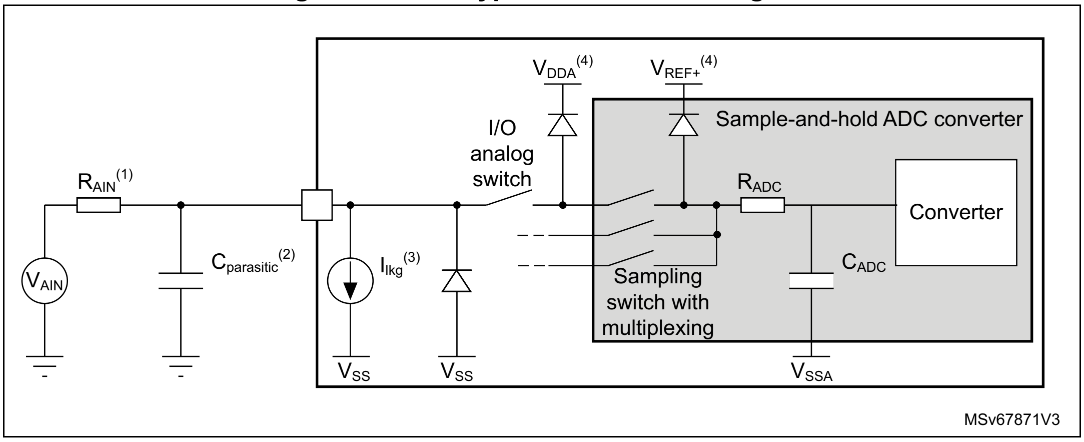
Figure 27. ADC typical connection diagram



1. Refer to *Table 56: ADC characteristics* for the values of R_{AIN} and C_{ADC}.
2. $C_{parasitic}$ represents the capacitance of the PCB (dependent on soldering and PCB layout quality) plus the pad capacitance (refer to *Table 51: I/O static characteristics* for the value of the pad capacitance). A high $C_{parasitic}$ value will downgrade conversion accuracy. To remedy this, f_{ADC} should be reduced.
3. Refer to *Table 51: I/O static characteristics* for the values of I_{lkg}.
4. Refer to *Figure 2: Power supply overview*.

General PCB design guidelines

Power supply decoupling should be performed as shown in *Figure 14: Power supply scheme*. The 100 nF capacitor should be ceramic (good quality) and it should be placed as close as possible to the chip.

5.3.17 Temperature sensor characteristics

Table 59. Temperature sensor characteristics

Symbol	Parameter	Min	Typ	Max	Unit
T_L(1)	V_{SENSE} linearity with temperature	-	±1	±5	°C
Avg_Slope(2)	Average slope from V_{SENSE} voltage	2.4	2.53	2.65	mV/°C
V_{30}(3)	Voltage at 30°C (±5 °C)	0.742	0.76	0.786	V
$t_{START(TS\_BUF)}$(1)	Sensor Buffer Start-up time in continuous mode	-	8	15	µs
t_{START}(1)	Start-up time when entering in continuous mode	-	70	120	µs
t_{S_temp}(1)	ADC sampling time when reading the temperature	5	-	-	µs
i_{sens}(1)	Temperature sensor consumption from V_{DD}, when selected by ADC	-	4.7	7.0	µA

1. Specified by design. Not tested in production.
2. Evaluated by characterization. Not tested in production.
3. Measured at V_{DDA} = 3.0 V ±10 mV. The V_{30} ADC conversion result is stored in the TS_CAL1 byte.

5.3.18 Timer characteristics

The parameters given in the following tables are specified by design.